MATH 127 Calculus for the Sciences

Lecture 26

Today's lecture

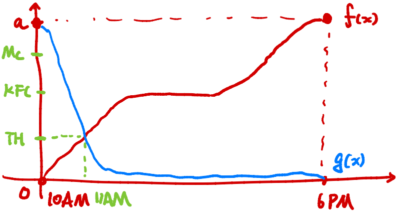
Last time
Critical points
Local extrema

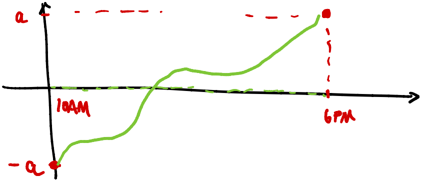
This time
Course note coverage Section 3.3.4
Intervals of Concavity
Inflection point

This super hard question from practice quiz

9. Suppose you leave for Montreal from Waterloo at 10 a.m. on Friday morning and arrive in Montreal at 6 p.m. The following Monday, you begin your return trip at 10 a.m. and arrive back in Waterloo at 6 p.m.

Show that at some point on the trip, there is a time where you will be at the exact same point on the road on both Monday and Friday.





We want to show

$f(x)=g(x)$

at some x.

This is the same as

$f(x)-g(x)=0$

Since $(f-g)(x)<0$ at 10AM

$(f-g)(x)>0$ at 6 PM

By IVT, there exist some

x such that $(f-g)(x)=0$

Recap

Condition	What we say
$f'(a) > 0$	f is increasing at $x = a$
$f'(a) < 0$	f is decreasing at $x = a$
$f'(a) = 0$	f has a critical point at $x = a$
$f'(a)$ does not exist	f has a critical point at $x = a$

Next, assume $f'(a) = 0$, then we have

$f'(x) < 0$ on the left of a $f'(x) > 0$ on the right of a	f is a local min at $x = a$
$f'(x) > 0$ on the left of a $f'(x) < 0$ on the right of a	f is a local max at $x = a$
$f(x)$ is a flat line near a	$x = a$ is both local max and local min

Concavity

The functions

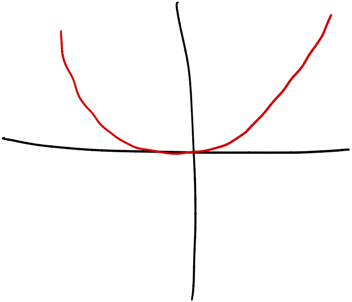
$$x^2, x^4, x^6$$

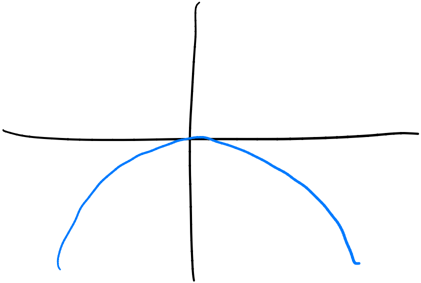
concave up, while the functions

$$-x^2, -x^4, -x^6$$

concave down.

This can be seen from the graphs. But how do we describe this behaviour in mathematics?



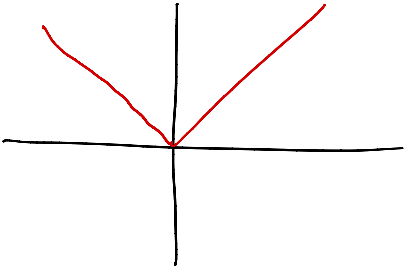


Concavity

Perhaps we can say the function concave up if the graph "hugs" toward the top. But consider

$$y = |x|$$

Is this function concave up? No

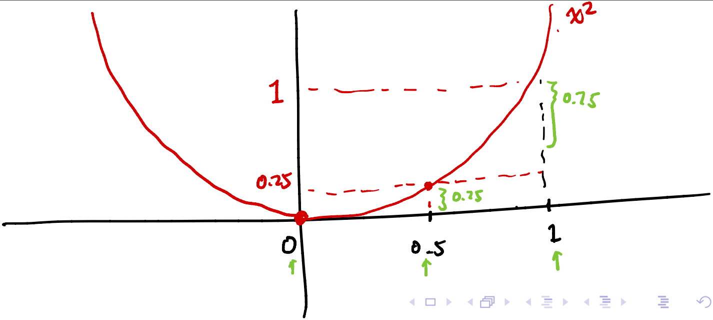


Concavity

We want x^2 to concave up, while $|x|$ should not concave up. Observe how x^2 increases.

$$\text{When the input is positive, the function } x^2 \text{ increases } \begin{cases} \boxed{\text{faster and faster?}} \\ \text{slower and slower?} \end{cases}$$

$$\text{When the input is negative, the function } x^2 \text{ decreases } \begin{cases} \text{faster and faster?} \\ \boxed{\text{slower and slower?}} \end{cases}$$



Concavity definition

When the input is negative, the function x^2 decreases slower and slower.

But saying decreasing slower is the same as increasing faster. They call that relativism.

Definition A function $f(x)$ is said to **concave up** if

$f'(x)$ is $\boxed{\text{increasing}}$, or equivalently, if $f''(x)$ is $\boxed{\text{positive}}$.

A function $f(x)$ is said to **concave down** if

$f'(x)$ is $\boxed{\text{decreasing}}$, or equivalently, if $f''(x)$ is $\boxed{\text{negative.}}$.

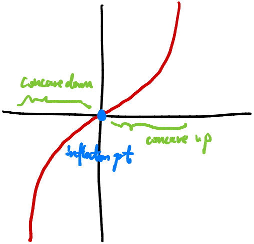
Remark Law of diminishing returns:
If you eat a lot of vitamin pills, your body will not be able to handle it. It will absorb slower and slower. So the more vitamin you eat, the slower your body absorbs it. So the graph of vitamin you absorb with respect to what you eat concaves $\boxed{\text{down}}$.

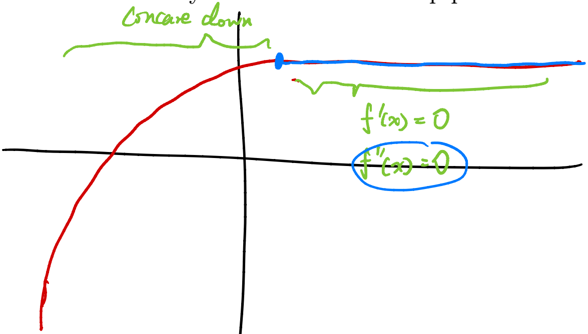
Inflection points

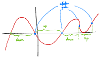
So if $f''(x) > 0$ you get concave up, and if $f''(x) < 0$ you get concave down. What do you get in between, when $f''(x) = 0$?

You get **Inflection points**.

In a graph, this just looks like the midway between a concave up part and a concave down part.



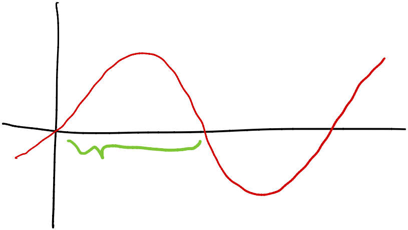




Example

Example Find the intervals of concavity, and inflection points of $f(x) = sin(x)$.

General tip: when you are asked to find intervals of whatever, and you convert that to a problem of finding signs of whatever, then you can find the zeroes first, then split then x-axis and draw a sign table.



1. $f'(x) = \cos(x)$

$f''(x) = -\sin(x)$

2. $f''(x) = 0$ if $-\sin(x) = 0$

$x = \dots, -2\pi, -\pi, 0, \pi, 2\pi, \dots$

or $x = k\pi$ for some integer k. ← inflection points.

	$-2\pi < x < -\pi$	$-\pi < x < 0$	$0 < x < \pi$	$\pi < x < 2\pi$
$\sin(x)$	+	−	+	−
$-\sin(x)$	−	+	−	+

3. In conclusion.

$\sin(x)$ is concave up on the intervals $((2k+1)\pi, (2k+2)\pi)$

concave down on $(2k\pi, (2k+1)\pi)$ for integer k.

Summary

Condition	What we say
$f'(a) > 0$	f is increasing at $x = a$
$f'(a) < 0$	f is decreasing at $x = a$
$f'(a) = 0$	f has a critical point at $x = a$
$f'(a)$ does not exist	f has a critical point at $x = a$
$f''(a) > 0$	f is concave up at $x = a$
$f''(a) < 0$	f is concave down at $x = a$
$f''(a) = 0$	f has an inflection point at $x = a$
$f''(a)$ does not exist	not much we can say

Next, assume $f'(a) = 0$, then we have

$f'(x) < 0$ on the left of a $f'(x) > 0$ on the right of a	f is a local min at $x = a$
$f'(x) > 0$ on the left of a $f'(x) < 0$ on the right of a	f is a local max at $x = a$
$f(x)$ is a flat line near a	$x = a$ is both local max and local min

Guess today's number

Survey